BUSINESS QUESTION AND VISUALIZATION REPORT

Project Detail	Information
Date	31 October 2026
Enrollment No.	EN25CS3010450
Project Name	Global Fuel Prices and Taxation Trends
Project Type	Individual Project
Maximum Marks	5 Marks
Student Name	Joyal Sengar

Visualization Development

Visualization development transforms the prepared global fuel dataset into graphical views that support comparison, exploration and interpretation. In this project, Tableau is used to examine fuel prices, taxation, regional differences, income and subsidy categories, price volatility, and the relationship between fuel prices and Brent crude measures.

Business Questions and Visualisation

The following business questions guide the Tableau worksheets. Each question is paired with the visualization used in the project and a concise observation based on the submitted screenshot.

1. How does the average fuel-tax percentage vary across regions and income levels?

Visualization: Highlight table / heatmap showing region against income level, with average fuel-tax percentage displayed in each cell.

Observation: The view allows comparison of tax percentages across region–income combinations. The displayed values vary by region and income group; blank cells indicate combinations for which no value is displayed in this view.

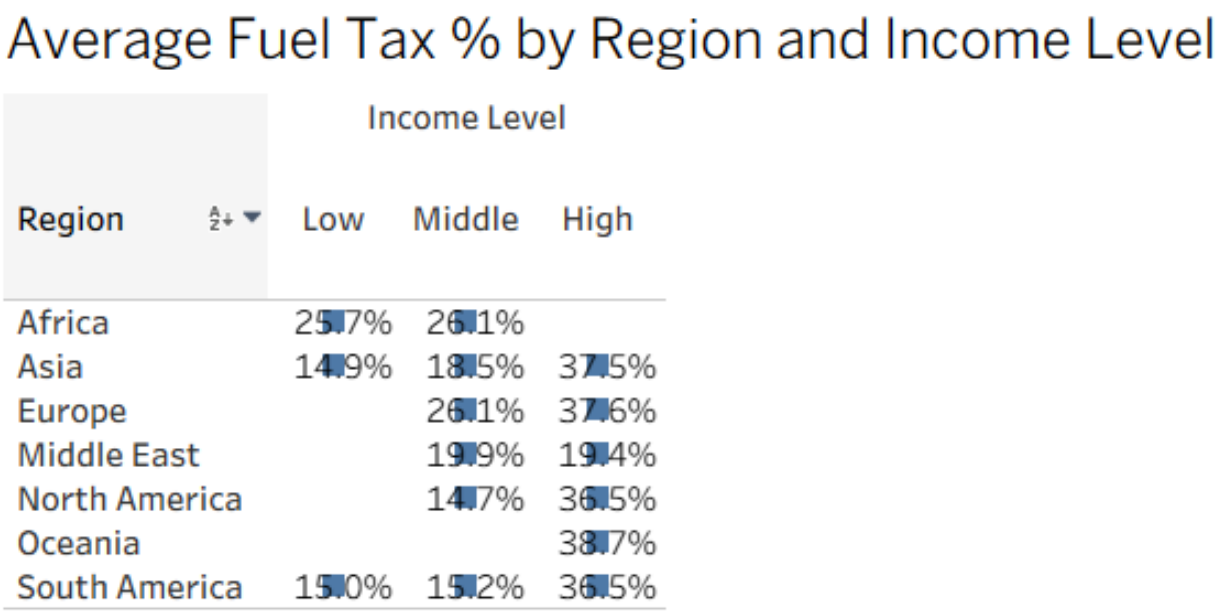


Figure 1: Average Fuel Tax % by Region and Income Level

2. How do average petrol-price values and average tax percentages differ across subsidy levels?

Visualization: Combined chart using circles for petrol-price values and a line for average tax percentage across subsidy levels.

Observation: The chart shows a high petrol-price value for the Low subsidy category, while the displayed average tax percentages for the categories are close to 15. The axes use different scales, so the two measures should be read independently.

Subsidy Level vs Average Petrol Price and Tax

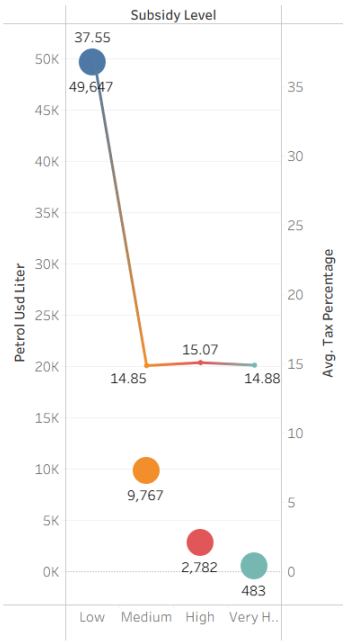


Figure 2: Subsidy Level vs Average Petrol Price and Tax

3. What relationship can be observed between tax percentage and average petrol price across countries?

Visualization: Scatter plot of tax percentage against average petrol price, with trend lines for the displayed groups.

Observation: The plotted points and trend lines show an upward pattern in the displayed groups. This indicates an association in this dataset view; it does not establish that tax percentage causes petrol prices to increase.

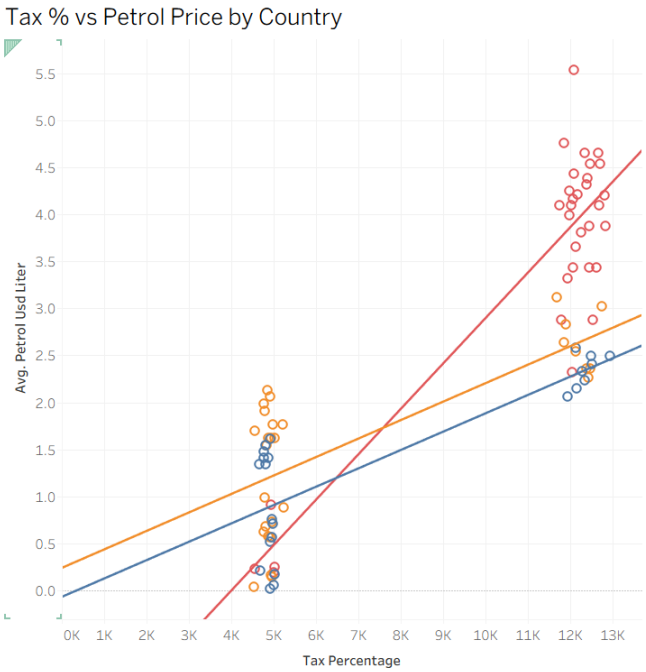


Figure 3: Tax % vs Petrol Price by Country

4. How have global fuel-price measures changed from 2020 to 2026, and what does the displayed 12-month forecast suggest?

Visualization: Multi-line time-series chart with a forecast region extending into 2027.

Observation: The historical lines show a decline around 2020, a rise into 2022, a subsequent correction, and later fluctuations through 2026. The shaded forecast area widens toward 2027, communicating increasing uncertainty. Forecast values should be treated as estimates, not actual observations.

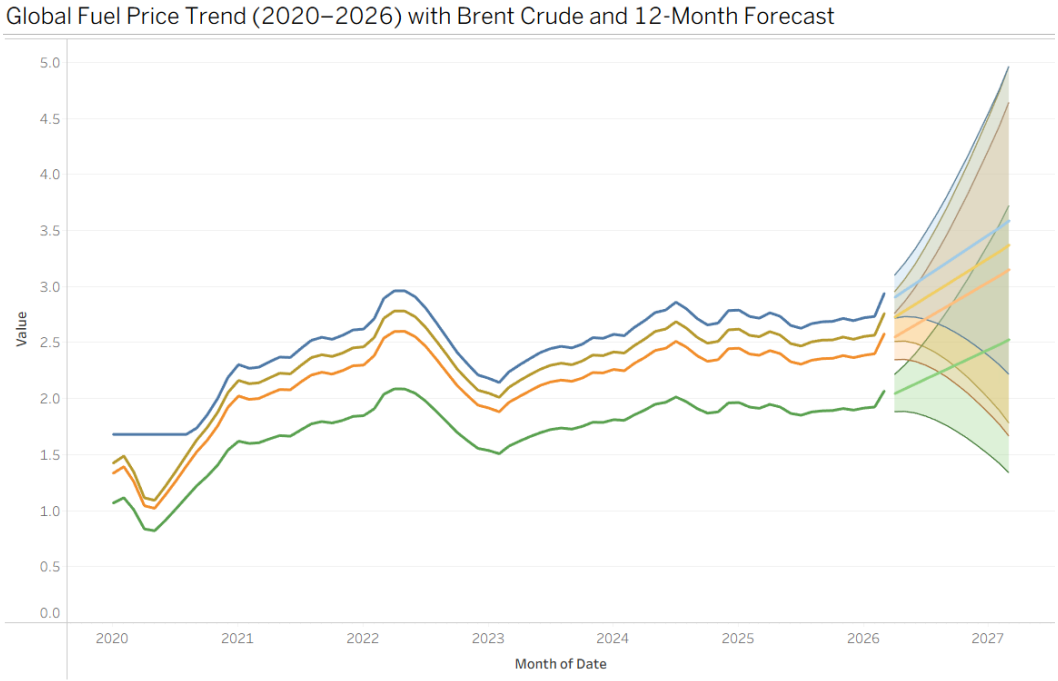


Figure 4: Global Fuel Price Trend (2020–2026) with Brent Crude and 12-Month Forecast

5. How does the average diesel price vary geographically across countries?

Visualization: Filled world map coloured by average diesel price (USD per litre).

Observation: The map provides a geographic comparison of average diesel prices. Countries are shaded differently, making regional variation visible; areas without a displayed value should not be interpreted as zero.

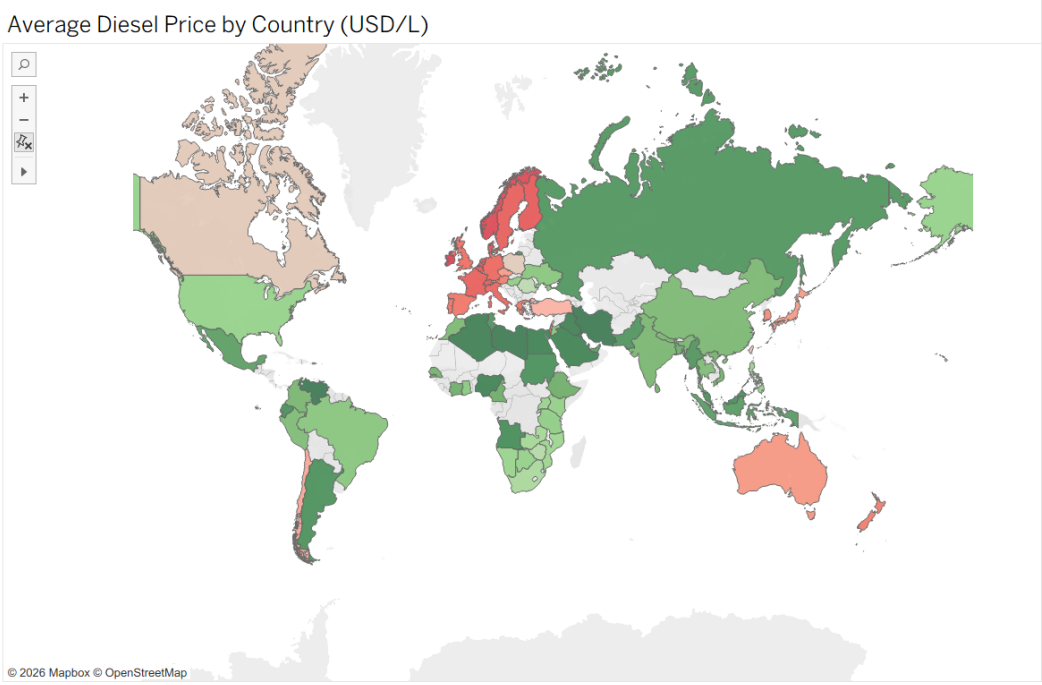


Figure 5: Average Diesel Price by Country (USD/L)

6. Which regions show the highest price volatility in the dataset?

Visualization: Horizontal bar chart ranking regions by price volatility.

Observation: South America has the longest bar in the displayed ranking, followed by the Middle East and Africa. Asia, North America, Oceania and Europe appear lower and relatively close together in this view.

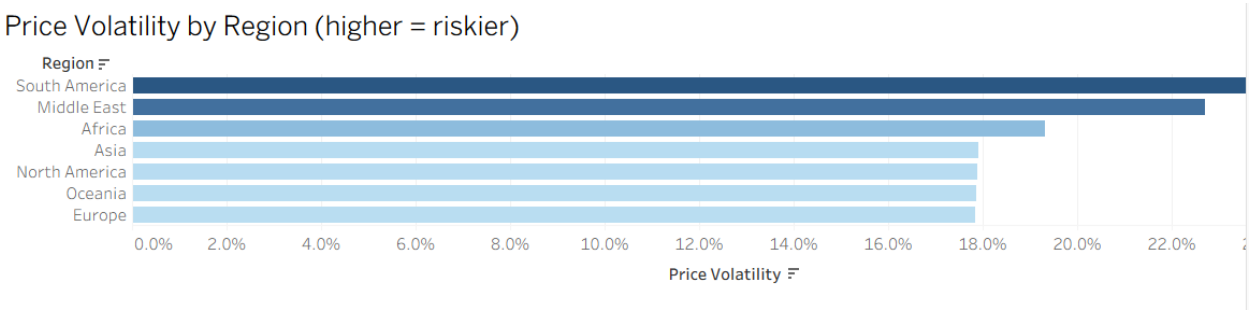


Figure 6: Price Volatility by Region (higher = riskier)

7. How do average diesel, LPG and petrol prices compare across income levels?

Visualization: Grouped bar chart comparing average diesel, LPG and petrol price measures for low-, middle- and high-income groups.

Observation: The displayed high-income group has visibly higher average values across the three fuel measures than the low- and middle-income groups. The chart labels show the values used in this comparison; interpretation should consider the dataset's units and coverage.

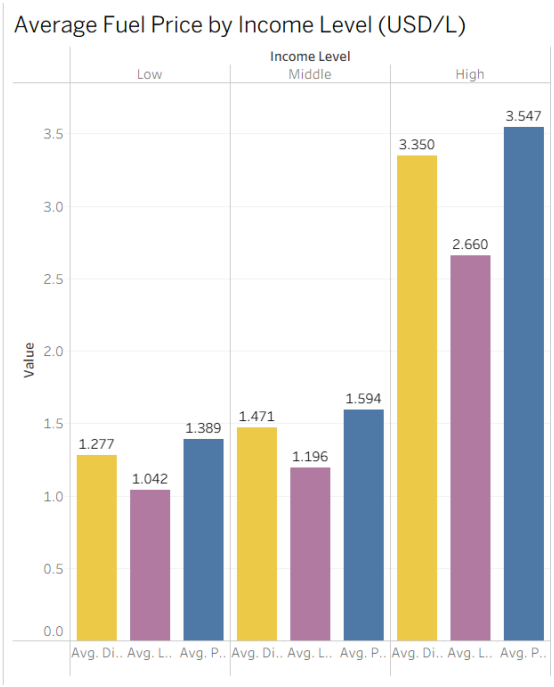


Figure 7: Average Fuel Price by Income Level (USD/L)

8. How does petrol price vary across income groups and subsidy levels?

Visualization: Highlight table / heatmap with income level as rows and subsidy level as columns.

Observation: The table makes it possible to compare petrol-price values across income and subsidy combinations. In the displayed cells, the Low-income/Low-subsidy and High-income/Low-subsidy combinations have comparatively high values; some combinations are blank.

Petrol Price by Income and Subsidy Level

Subsidy Level

Income L..	Low	Medium	High	Very H..
Low	2.348	1.449	0.640	0.114
Middle	2.642	1.793	0.719	0.115
High	3.997		0.914	0.225

Figure 8: Petrol Price by Income and Subsidy Level

9. How does average Brent crude price relate to average diesel price across income groups?

Visualization: Scatter plot of average Brent crude price (USD) against average diesel price (USD/litre), with separate trend lines for the displayed income groups.

Observation: The plotted points and trend lines show an upward relationship between Brent crude and diesel-price measures in the displayed groups. This is an observed association in the dataset and should not be interpreted as proof of causation.

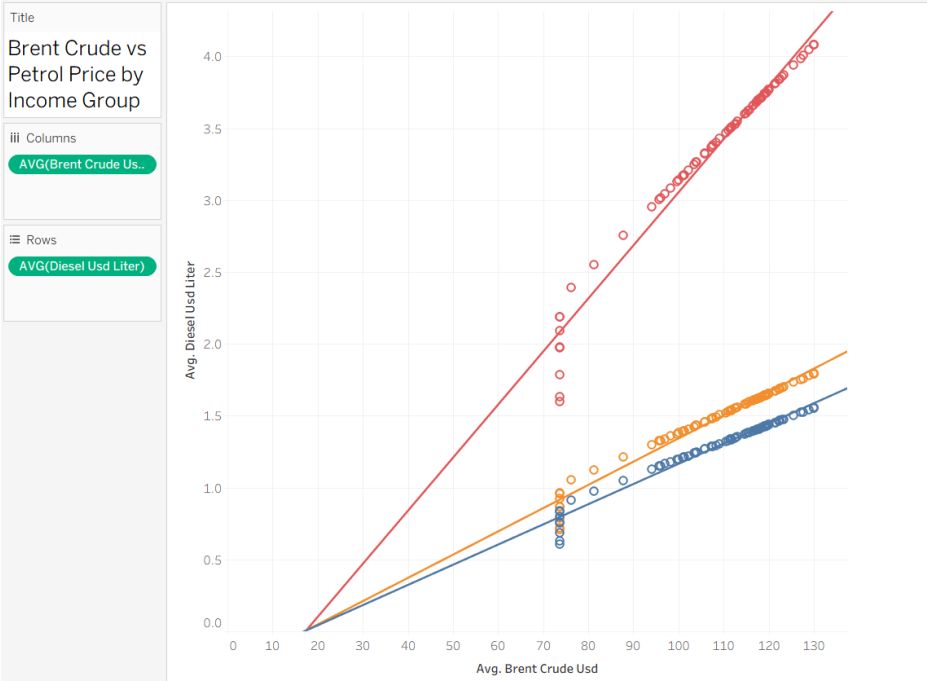


Figure 9: Brent Crude vs Diesel Price by Income Group

Overall Summary

The nine Tableau visualizations provide complementary views of global fuel pricing and taxation: regional and income-based comparisons, subsidy analysis, country-level price patterns, historical trends and forecast, volatility ranking, and comparisons with Brent crude. Together, they help users explore patterns in the available dataset. The visualizations are descriptive; observed relationships should not be treated as proof of cause and effect.